

Week 6 Assignment: Module Six Activity Reflection

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Design, Interactivity, and Accessibility in a COVID-19 Visualization Dashboard

This reflection accompanies the Module Six Activity dashboard, which visualizes cumulative COVID-19 death data from Our World in Data across 246 countries and aggregate regions. The discussion below covers the design choices that drove the visualizations, the technical challenges encountered with interactivity and animation, and the accessibility and cognitive-load principles built into the final product.

Design Choices and Their Contribution to Clarity

Several deliberate choices shaped the dashboard's clarity and impact. Color palette selection was the first. I used the viridis colormap for the static correlation heatmap and the top-20 bar chart because viridis is perceptually uniform, meaning that equal steps in the underlying values produce equal-seeming steps in brightness. It also remains distinguishable under the three most common forms of color vision deficiency. For the Plotly interactive charts, I switched to the Set2 qualitative palette because categorical country comparisons need contrasting hues rather than a sequential gradient.

Hierarchy and filtering shaped what the viewer actually sees. The raw dataset mixes aggregate regions such as World, Europe, and High-income countries into the same columns as individual nations. Including those in a top-countries chart would have crowded out actual country comparisons because the aggregates dwarf any single country. I built an explicit

exclusion list and applied it consistently across Parts 2, 3, and 4 so every ranked visualization compared like with like.

The inclusion of strategic annotations shifts the chart from data visualization to data storytelling. Labeling the one-million cumulative death milestone alongside the current total establishes vital temporal landmarks, offering viewers a cognitive framework that an unannotated trend line lacks. A subtler choice was preserving magnitude where it mattered. Part 1 normalizes every country column to a 0–1 range, which is appropriate for correlation analysis. For the bar chart and the global time series, however, I deliberately reloaded the un-normalized data so the y-axis values would read as real death counts. A top-20 chart where every bar reaches 1.0 would have communicated nothing about scale.

Challenges Implementing Interactivity and Animation

To overcome data preprocessing and rendering challenges, I implemented four key technical solutions. First, because `MinMaxScaler` erased relative magnitudes by capping all country maximums at 1.0, I loaded the original CSV inside specific visualization functions to restore accurate rankings while maintaining the required preprocessed workflow. Second, to fix sluggish performance and massive file sizes caused by animating 1,674 daily frames, I resampled the data to a weekly frequency, reducing the animation to a smooth ~240 frames. Third, to ensure the Matplotlib animation survived the HTML export, I wrapped it in `HTML(ani.to_jshtml())` to embed a functional JavaScript-driven player. Finally, to resolve visual confusion from Plotly applying a range slider across stacked subplots, I explicitly scoped the slider to the detail subplot

so the overview remained static while the detail pane retained its interactivity.

Accessibility and Cognitive-Load Considerations

To optimize accessibility and minimize cognitive load, I focused on visual clarity, data restraint, and progressive disclosure. I ensured all color palettes are colorblind-safe and styled all charts with clean sans-serif typography, readable sizing, explicit labeling, and comma-formatted numbers for instant legibility. To keep the visualizations digestible, I restricted the data to small, focused subsets of top-performing countries and stripped away distracting chart elements like unneeded spines and heavy gridlines. Furthermore, I managed interactive complexity by implementing named time-range selectors rather than open-ended zooming, using a dropdown filter to isolate single countries, and splitting the global and country-level trends into a unified Overview + Detail chart. Together, these design choices ensure the data is immediately understandable at a glance while remaining seamlessly explorable for deeper analysis.

AI Usage Acknowledgment

Anthropic's Claude (claude.ai) was used to assist with Python code structure and errors. All analysis, interpretation, and editorial decisions are my own.